

An Artifact for the Article:

“VerLog: Enhancing Release Note Generation for Android Apps using Large Language Models”

Abstract

This artifact contains the implementation of VerLog, a tool that enhances release note generation for Android applications using Large Language Models. The package includes VerLog's source code, compiled executable JAR files, execution scripts, metadata of benchmark apps used in our evaluation, and multiple versions of compiled APK files and code repository of PairDrop[1], a popular open-source app used as an example input. All materials are available on Zenodo[2].

The artifact expects to demonstrate how VerLog generates high-quality release notes that outperform existing tools when provided with code repositories and compiled APK files of base and release versions, corresponding to RQ1 in our paper. For convenient evaluation, we prepared PairDrop as input for demonstration. By running the execution script, evaluators can observe the complete pipeline from code change extraction to release note generation, as every intermediate output will be saved. The PairDrop example showcases VerLog's performance across multiple version pairs, with results organized in structured output directories.

VerLog requires Java ≥ 1.8 , Python ≥ 3.7 , and Android platforms. For ease of evaluation, we provide a Docker image[3] with all dependencies pre-installed. Evaluators only need to configure their LLM API key (demonstrated with DeepSeek for cost efficiency). The artifact's modular design allows integration with different LLM providers and extension to programming languages beyond Java, making it valuable for both research and practical applications.

[1] <https://github.com/fm-sys/snapdrop-android>

[2] <https://zenodo.org/records/15200248>

[3] <https://hub.docker.com/r/jarweigh/verlog-artifact>